

Dataset Reference — All 10 Candidate DNN Directions

1. Fusion robust to missing modality (clinical image/signal + tabular)

Dataset	Source	Size	License	Access
ISIC 2019	https://challenge.isic-archive.com/data/#2019	25,331 dermoscopic images, 8 classes	CC BY-NC 4.0	Free account on ISIC archive; also mirrored on Kaggle (e.g. andrewmvd/isic-2019)
HAM10000	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T	10,015 dermoscopic images, 7 classes	CC BY-NC 4.0	Direct download from Harvard Dataverse; also on Kaggle (kmader/skin-cancer-mnist-ham10000)
PTB-XL	https://physionet.org/content/ptb-xl/1.0.3/	21,837 12-lead ECGs, 18,885 patients	CC BY 4.0	Open access tier — no PhysioNet credentialing needed. Direct <code>wget</code> from PhysioNet.

2. Multimodal emotion recognition — CMU-MOSEI

Dataset	Source	Size	License	Access
CMU-MOSEI	https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK	23,453 utterances, 3,228 videos, 65 hrs	Research use (CMU MultiComp terms)	Install <code>CMU-MultimodalSDK</code> (pip/git), it fetches precomputed text/audio/visual features
MELD (cross-dataset)	https://affective-meld.github.io/	13,708 utterances	Research use	Direct download from project site
IEMOCAP (cross-dataset)	https://sail.usc.edu/iemocap/	12 hrs, 10 actors	Research use, requires signed license form	Request access via USC SAIL form

3. Self-supervised pretraining for small medical signal data (EEG/ECG)

Dataset	Source	Size	License	Access
PTB-XL	(same as above) https://physionet.org/content/ptb-xl/1.0.3/	21,837 ECGs	CC BY 4.0	Open, no credentialing

Dataset	Source	Size	License	Access
OpenNeuro (EEG)	https://openneuro.org/	varies by dataset	Mostly CC0/PDDL	Free, browse and download per-dataset, no registration for most
TUH EEG Corpus	https://isip.piconepress.com/projects/tuh_eeg/	20,000+ EEG recordings	Free but requires registration form	Sign up at Temple University Neural Engineering Data Consortium

4. Wearable sensor fusion for human activity recognition

Dataset	Source	Size	License	Access
PAMAP2	https://archive.ics.uci.edu/dataset/231/pamap2+physical+activity+monitoring	18 activities, 9 subjects, 3 IMUs + HR, 100Hz	CC BY 4.0	Direct download, UCI ML Repository, no signup
UCI-HAR	https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones	10,299 samples, 30 subjects	CC BY 4.0	Direct download, no signup
WISDM	https://www.cis.fordham.edu/wisdm/dataset.php	~1M accelerometer readings	Free research use	Direct download from Fordham WISDM lab page
OPPORTUNITY	https://archive.ics.uci.edu/dataset/226/opportunity+activity+recognition	6 subjects, dense sensor rig	CC BY 4.0	Direct download, UCI ML Repository, no signup

5. Graph neural networks for molecular property prediction

Dataset	Source	Size	License	Access
MoleculeNet (BBBP, Tox21, SIDER, ClinTox, ESOL, FreeSolv, Lipophilicity, BACE)	https://moleculenet.org/datasets-1	ranges from ~600 to ~93K molecules per split	Mostly public domain / open	Fetches programmatically via <code>deepchem.molnet</code> or direct CSV download
OGB graph-property datasets	https://ogb.stanford.edu/docs/graphprop/	e.g. ogbg-molhiv: 41K graphs	ODC-BY / open	<code>pip install ogb</code> , auto-downloads via <code>PygGraphPropPredDataset</code>

6. Self-supervised, multimodal remote sensing

Dataset	Source	Size	License	Access
EuroSAT	https://github.com/phelber/EuroSAT	27,000 64×64 Sentinel-2 patches, 10 classes	Free for research (MIT-style on repo)	Direct download link in repo README; also on Kaggle (apollo2506/eurosat-dataset)
BigEarthNet-MM	https://bigearth.net/	~590K paired Sentinel-1/2 patches (full); use a subset	CC0 / open, no signup	Direct download from bigearth.net (large — subsample)
SSL4EO-S12	https://github.com/zhu-xlab/SSL4EO-S12	~1M unlabeled Sentinel-1/2 patch series	Open (CC BY 4.0)	Download links + scripts in repo

7. Environmental sound classification (fusion or compression)

Dataset	Source	Size	License	Access
ESC-50	https://github.com/karolpiczak/ESC-50	2,000 clips, 50 classes	CC BY-NC 3.0	Direct git clone / zip download, no signup
ESC-10	(subset of ESC-50, same repo)	400 clips, 10 classes	CC BY-NC 3.0	Same as ESC-50
UrbanSound8K	https://urbansounddataset.weebly.com/urbansound8k.html	8,732 clips, 10 classes	CC BY-NC 3.0	Free download after short request form (auto-approved)

8. Model compression for edge deployment

Dataset	Source	Size	License	Access
CIFAR-10/100	https://www.cs.toronto.edu/~kriz/cifar.html	60,000 32×32 images	Free for research	Direct download, no signup
ESC-50 / PAMAP2	(see #4, #7 above)	—	—	—

9. Network traffic anomaly detection with self-supervised pretraining

Dataset	Source	Size	License	Access
CIC-IDS2017	https://www.unb.ca/cic/datasets/ids-2017.html	2,830,743 flows, ~80 features, 8 CSVs	Free for research	Direct download from UNB CIC site, no signup (large files)

Dataset	Source	Size	License	Access
CSE-CIC-IDS2018	https://www.unb.ca/cic/datasets/ids-2018.html	16.2M+ flows, 75 features	Free for research	Direct download / AWS S3 bucket instructions on same page (very large — subsample)

10. Energy load forecasting with hybrid architecture

Dataset	Source	Size	License	Access
UCI Household Electric Power Consumption	https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption	2,075,259 minute-level readings	CC BY 4.0	Direct download, no signup
ETT family (ETTh1/h2, ETTm1/m2)	https://github.com/zhouhaoyi/ETDataset	2 years, hourly/15-min	Public/research use	Direct git clone / CSV download
PJM Hourly Energy Consumption	https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption	~10 years hourly, multiple regions	Public domain (PJM data)	Kaggle download (needs Kaggle account)
ISO-NE Load Data	https://www.iso-ne.com/isoexpress/web/reports/load-and-demand	Historical hourly load	Free, public	Direct download from ISO-NE reports portal
Open-Meteo (weather covariates)	https://open-meteo.com/	API, historical weather	Free, no key required for non-commercial	REST API, no signup for reasonable use

Notes / gotchas carried over from the memo

- **Skip MIMIC-IV, MIMIC-CXR, BRSET** — PhysioNet credentialing (CITI training + data use agreement + identity review) can take days to 45 days. None of the datasets above require this; PTB-XL is explicitly the open-access tier.
- **ISIC / HAM10000 are CC BY-NC** — fine for coursework, not for anything commercial.
- **CIC-IDS2018 and full BigEarthNet are large** — plan to subsample or stream; likely to exceed free-tier Colab/Kaggle RAM if loaded whole.
- **UrbanSound8K** requires filling a short form on the Weebly page before the download link is emailed/shown — it's free and near-instant, not a credentialing process.